Started on	2023-10-09 07:48
State	Finished
Completed on	2023-10-09 08:59
Time taken	1 hour 10 mins
Grade	**9.00** out of 10.00 (**90**%)

Question **1**
Correct
Mark 1.00 out of 1.00

Choose the second order non-homogeneous linear differential equation with constant coefficients from the list below.

Select one:

- $y'' + 2y' - y = 0$
- $y'' + 2y' - y = \sin t$ ✓
- $y'' + 2(y')^2 - y = 0$
- $y'' + 2ty' - y = 0$

Your answer is correct.

The correct answer is: $y'' + 2y' - y = \sin t$

Question **2**
Correct
Mark 1.00 out of 1.00

Choose the equation for which the method of undetermined coefficients can be applied to find a particular solution.

Select one:

- $y'' + 2y' - y = t^2 \cos t$ ✓
- $y'' - 4y' + 3y = \frac{1}{t}$
- $y'' + 3y' - 6y = \tan t$
- $y'' + 6y' - 2y = \sin(e^t)$

Your answer is correct.

The correct answer is: $y'' + 2y' - y = t^2 \cos t$

Question 3
Incorrect
Mark 0.00 out of 1.00

Choose the equation for which the method of undetermined coefficients can NOT be applied to find a particular solution.

Select one:

- $y'' + 2y' - y = \frac{1}{t}$
- $y'' - 4y' + 3y = (5t^2 + 4t)\sin(3t)$
- $y'' + 3y' - 6y = (4t - 1)\cos t$
- $y'' + 6y' - 2y = e^{-t}\cos(5t)$ ✗

Your answer is incorrect.

The correct answer is: $y'' + 2y' - y = \frac{1}{t}$

Question 4
Correct
Mark 1.00 out of 1.00

Find a particular solution of $y'' + 3y' + 2y = 3t$ using undetermined coefficients method.

Select one:

- $y = \frac{3}{2}t - \frac{9}{4}$ ✓
- $y = \frac{3}{4}t + \frac{9}{2}$
- $y = \frac{3}{2}t^2 - \frac{9}{4}t$
- $y = \frac{3}{2}t + \frac{9}{4}$

Your answer is correct.

The correct answer is: $y = \frac{3}{2}t - \frac{9}{4}$

Question 5
Correct
Mark 1.00 out of 1.00

Find a particular solution of $y'' + 3y' + 2y = 10e^{3t}$ using undetermined coefficients method.

Select one:

- $y = \frac{e^{3t}}{2}$ ✓
- $y = \frac{e^{-3t}}{2}$
- $y = \frac{e^{3t}}{4}$
- $y = \frac{e^{-3t}}{4}$

Your answer is correct.

The correct answer is: $y = \frac{e^{3t}}{2}$

Question 6
Correct
Mark 1.00 out of 1.00

Find a particular solution of $y'' + 3y' + 2y = \sin t$ using undetermined coefficients method.

Select one:

- $y = \frac{\sin t}{10} - \frac{3\cos t}{10}$ ✓
- $y = \frac{\sin t}{10} + \frac{3\cos t}{10}$
- $y = \frac{3\sin t}{10} - \frac{\cos t}{10}$
- $y = \frac{3\sin t}{10} + \frac{\cos t}{10}$

Your answer is correct.

The correct answer is: $y = \frac{\sin t}{10} - \frac{3\cos t}{10}$

Question 7
Correct
Mark 2.00 out of 2.00

Find a particular solution of $y'' - 3y' - 4y = 3e^{2t} + 2\sin t$ using undetermined coefficients.

Select one:

- $y_p = -\frac{e^{2t}}{2} + \frac{3}{17}\cos t - \frac{5}{17}\sin t$ ✓
- $y_p = -\frac{e^{2t}}{4} + \frac{3}{17}\cos(2t) - \frac{5}{17}\sin(2t)$
- $y_p = -\frac{e^{2t}}{2} + \frac{3}{17}\cos(2t) - \frac{5}{17}\sin(2t)$
- $y_p = -\frac{e^{2t}}{2} - \frac{3}{17}\cos t + \frac{5}{17}\sin t$

Your answer is correct.

The correct answer is: $y_p = -\frac{e^{2t}}{2} + \frac{3}{17}\cos t - \frac{5}{17}\sin t$

Question 8
Correct
Mark 2.00 out of 2.00

Find a particular solution of $y'' + 3y' + 2y = \frac{1}{1+e^t}$ using variation of parameters method.

Select one:

- $y_p = (\ln(1+e^t))e^{-t} + (\ln(1+e^t) - e^t)e^{-2t}$ ✓
- $y_p = (\ln(1+e^t))e^{t} + (\ln(1+e^{-t}) - e^{-t})e^{-2t}$
- $y_p = (\ln(1+e^{-2t}))e^{-t} + (\ln(1+e^t) - e^t)e^{-2t}$
- $y_p = (\ln(1+e^{2t}))e^{-t} + (\ln(1+e^t) - e^{-2t})e^{-2t}$

Your answer is correct.

The correct answer is:
$y_p = (\ln(1+e^t))e^{-t} + (\ln(1+e^t) - e^t)e^{-2t}$